

RESEARCH ARTICLE

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Movement-related potentials associated with movement preparation and motor imagery

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Abstract Movement-related potentials (MRPs), reflecting cortical activity associated with voluntary movement, typically show a slowly increasing negative potential beginning between 1 and 2 s prior to movement, which most likely reflects motor preparatory processes. Studies of regional cerebral blood flow implicate the supplementary motor area in such preparatory processes; however, the contribution of the supplementary motor area to pre-movement activity observed in MRPs is debated. It is possible to examine MRPs relating to movement preparation alone, in the absence of movement execution, by recording MRPs associated with imagined movements. In this study, MRPs were recorded from 11 healthy control subjects while performing a sequential button-pressing task in response to external cues, and while imagining performance of the same task in response to the same cues. The early component of MRPs was found not to differ in amplitude, onset time, or topography when performing compared with imagining movement, indicating that both movement execution and motor imagery involve similar pre-movement preparatory processes generated in the same cortical area – most likely the supplementary motor area. It is therefore concluded that the early component of the MRP reflects activity arising predominantly from the supplementary motor area and is associated with pre-movement motor preparatory processes which occur relatively independently of actual movement execution.

Key words Movement-related potentials · Motor imagery · Supplementary motor area · Human

Introduction

Activity of the cortex associated with the preparation and execution of voluntary movement may be examined by averaging electroencephalographic (EEG) activity to reveal movement-related potentials associated with different aspects of movement performance. Movement-related potentials preceding voluntary movement consist of two major components. A late component usually begins within 500 ms prior to movement onset and is lateralised with a maximal amplitude recorded over the primary motor area of the contralateral hand during finger movements (Deecke et al. 1969; Shibasaki et al. 1980, Tamas and Shibasaki 1985). The late component therefore appears to reflect cortical activity associated with movement execution, arising predominantly from the contralateral primary motor area.

The early component generally begins between 1 and 2 s prior to movement onset and is bilaterally symmetrical across the scalp, with a maximal amplitude at the vertex (position Cz) (Shibasaki et al. 1980; Tamas and Shibasaki 1985; Deecke 1987). The early component may therefore reflect activity of the supplementary motor area which lies close to the vertex (Deecke 1987).

There is considerable evidence that the early component of the MRP reflects predominantly pre-movement activity of the supplementary motor area. In monkeys, single-cell recordings have found that neurones within the supplementary motor area show preparatory activity, often beginning between 1 and 2 s prior to movement (Romo and Schultz 1992; Tanji and Shima 1994), whereas neurones within primary motor cortex are more active in close temporal association with movement execution (Tanji and Shima 1994).

In humans, however, the contribution of the supplementary motor area to MRPs is less clear. Recordings from subdural electrodes implanted within premotor and sensorimotor areas indicate that both supplementary and primary motor areas may contribute to early and late phases of pre-movement activity (Neshige et al. 1988;

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Ikeda et al. 1992), suggesting some functional overlap between these areas.

More recently, the generators of various components of MRPs have been investigated using dipole analysis methods, whereby hypothesised dipole sources are mathematically modelled to determine the sources which best account for spatio-temporal characteristics of observed MRPs. Such studies have shown mixed results, with some supporting a source located in the supplementary motor area (Toro et al. 1993; Tarkka 1994) and others not (Bötzel et al. 1993; Böcker et al. 1994). These studies, however, examined only simple *repetitive* finger movements, which may not have involved the supplementary motor area, since the supplementary motor area has been shown to play a greater role in complex and *sequential* movements (Benecke et al. 1985; Lang et al. 1988).

Studies of regional cerebral blood flow, which allow more accurate localisation of activity, consistently show increased activation of both the supplementary motor area and primary motor cortex during voluntary movement (Lassen et al. 1978; Roland et al. 1980; Shibasaki et al. 1993), as well as task-dependent changes in activity in prefrontal areas (Deiber et al. 1991). However, such techniques provide poor temporal discrimination; therefore, the time at which the supplementary motor area is active cannot be determined, and hence the stage of motor control processes to which it contributes is unclear. On the other hand, studies of MRPs provide good temporal resolution, but potentials are widespread and difficult to localise. Further, early and late components of MRPs, which are generally defined by differences in topographic characteristics, are difficult to separate, since they overlap in time.

It may be possible to separate motor preparatory and execution-related components by examining MRPs associated with imagined movements, reflecting movement preparation alone, in the absence of movement execution. The processes of motor imagery and execution share several similarities. Imagined and executed movements are both of similar duration (Decety and Michel 1989) and both result in autonomic activation involving increased heart rate and respiration (Decety et al. 1991, 1993).

Studies of regional cerebral blood flow have examined cortical activity associated with imagined and executed movements. Roland et al. (1980) found increased activation of supplementary motor, contralateral primary motor, and somatosensory areas during execution of a sequential finger-movement task. During imagined performance of the same task, however, increased activation was found only in the supplementary motor area, with no involvement of primary motor or somatosensory areas.

More recent studies have similarly shown strong activation of the supplementary motor area and not primary motor cortex during imagined movements (Decety et al. 1988; Rao et al. 1993). Less consistently, activation has also been reported in prefrontal areas (Decety et al. 1988), including dorsolateral prefrontal (Ceballos-Bau-

mann et al. 1994) and anterior cingulate areas (Stephan et al. 1995) during both imagined and executed movements. However, the main difference consistently reported is the lack of activity in primary motor areas and persistent activity in premotor and supplementary motor areas during imagined movement.

Therefore, the contribution of the supplementary motor area to pre-movement activity, independent of any contribution of primary motor areas, may be determined by examining potentials associated with imagined movements. Such studies, however, have been limited, examining potentials only *during* imagined and executed hand movements and not examining characteristics of the potential *prior* to movement (Naito and Matsumura 1994; Beisteiner et al. 1995). Thus, the temporal characteristics of *pre-movement* preparatory activity associated with imagined movements have not been previously examined.

The aim of this study, therefore, was to examine temporal characteristics of MRPs associated with imagined movements compared with executed movements, and to determine the role of the supplementary motor area in any observed preparatory activity. Subjects performed four experimental tasks, involving: (1) performing a sequential button-pressing task in response to external cues; (2) imagining performance of the same task in response to the same cues; (3) watching and focusing attention on the same sequence of external cues, but not imagining movement; and (4) fixating on a central point without attending to the ongoing cues.

It was essential to provide external cues to which MRPs could be time-locked, since subjects made no *overt* responses when imagining movement. However, the presence of such cues may result in cognitive potentials related to anticipation or expectancy, such as the contingent negative variation (CNV) (Walter et al. 1964). This was controlled by also examining potentials for non-movement-related conditions which involved the same anticipation and expectancy of regular cues, allowing the presence of any CNV to be determined.

Temporal characteristics of MRPs associated with each of the experimental tasks were examined by comparing the onset time and mean slope of the early component, and peak amplitude associated with the late component of potentials recorded from position Cz (lying close to the supplementary motor area). Subsequently, in a localisation phase of the study, the spatial distribution of potentials across various scalp locations was examined to determine any topographic differences in early and late components of potentials associated with motor imagery compared with movement execution.

Materials and methods

Subjects

Twelve healthy adults took part in the experiment; however results of only 11 subjects were included in the analysis (five men, six women) aged 19–55 years (mean 34.9 years). One subject seemed

unable to adequately switch between the different cognitive requirements of imagining movement and watching lights conditions and was therefore omitted from the sample. A subset of 6 of these subjects (mean age 37.2 years) also took part in the localisation phase of the study. All subjects were without previous history of stroke, serious head injury or other neurological disturbance, and all gave informed consent prior to participation in the study. Experimental work was carried out under the approval of local ethics committees in accordance with the declaration of Helsinki.

Tasks

Movement and event-related cortical potentials were recorded from the scalp during various experimental tasks. These tasks were performed on a tapping board (Mattingley et al. 1992), which consisted of a box containing two parallel rows of ten buttons, one above the other, with two centred start buttons and one centred end button (Fig. 1). Light-emitting diodes underneath each button could be illuminated and extinguished according to pre-arranged spatial and temporal patterns. For all experimental conditions, a pathway along the tapping board, consisting of a particular combination of ten top- and bottom-row buttons, was initially illuminated (Fig. 1). During experimental trials, lights underneath each button in the pathway were successively extinguished from right to left at a constant rate of one every 4 s.

Each subject performed four experimental tasks:

1. *Performing movement.* Subjects held down each illuminated button until the light underneath was extinguished, then moved as quickly as possible to the next illuminated button in the sequence (immediately to the left). Movements were always performed with the right hand.
2. *Imagining movement.* Subjects were instructed to *imagine* their finger holding down each illuminated button and, when the light underneath was extinguished, to imagine moving to press the next illuminated button in the sequence with the right hand, without actually performing the movement.
3. *Watching lights.* Moving their eyes only, subjects focused their attention on the last illuminated button to the right until the light underneath was extinguished, then focused on the next illuminated button in the sequence.
4. *Fixating on centre.* Subjects fixated on a marked point in the centre of the tapping board without moving their eyes, nor attending to cues, as lights were progressively extinguished from right to left. This condition enabled visual evoked potentials associated only with the light cue to be measured.

The order of conditions was counter-balanced such that half of the subjects performed conditions in the above order (performing movement, imagining movement, watching lights, fixating on centre) and half performed the reverse order. These orders involved systematically increasing or decreasing involvement of motor and cognitive processes.

Data acquisition

Movement-related potentials were recorded using an Amlab workstation (Associative Measurement, Sydney, Australia) which performed digital on-line processing and averaging of EEG and electromyographic (EMG) activity. The EEG was recorded from sil-

ver/silver chloride surface electrodes, with one recording electrode lying close to the supplementary motor area at position Cz, referenced bilaterally to electrodes over both mastoid processes, and with a ground electrode on the forehead. In the localisation phase of the study, nine recording electrodes were attached to the scalp at positions F3, Fz, F4, C3, Cz, C4, P3, Pz and P4 (10–20 system). EMG activity associated with upper limb movement was recorded from two gold-plated silver-cup electrodes separated by 4 cm and lying over the biceps brachii muscle of the right arm. Recording sites were first prepared with an abrasive conductive solution, and electrodes were attached using electrode cream. Electrode impedances were always below 5 k Ω .

The EEG was amplified using isolated AC amplifiers (gain 20 000 V/V, time constant 25 000 ms), digitised at a sampling rate of 100 Hz and filtered at 20 Hz (low-pass). The EMG was amplified at a gain of 10 000 V/V (time constant 97 ms), digitised at 100 Hz and filtered at 48 Hz (low-pass). EEG and EMG potentials were averaged over 4-s sweeps for successive events, time-locked to the extinction of light cues on the tapping board. Averaged potentials consisted of the period from 3 s before to 1 s after the cue. An artefact rejection system disabled the averager for any sweeps in which the recorded EEG potential deviated by more than 150 μ V peak-to-peak. Consequently, sweeps containing artefacts from vertical eye movements, blinks, and large EMG responses from neck and jaw movements were rejected from the mean. At least 100 sweeps were averaged in each condition for each subject. A square pulse of 10 μ V and 100 ms duration generated for each sweep allowed the subsequent calibration of averaged potentials.

Data analysis

Characteristics of averaged potentials for each subject in each condition were quantified by measures of *early slope*, *onset time*, and *peak amplitude*. The early slope consisted of the mean slope of the potential over the period from 1500 to 500 ms prior to the cue, calculated by linear regression. For movement-related conditions, the mean slope over this interval reflects the increase in neural activity associated with the early component of the MRP (Cunnington et al. 1995).

The onset time consisted of the point in time, prior to the cue, when the early component of the potential began to rise (Cunnington et al. 1995). Using iterative curve-fitting software, the potential over the period from 3000 to 500 ms prior to the cue was fitted with two linear segments representing: (1) baseline activity before onset of the MRP; and (2) the increasing potential of the early component of the MRP. The point of intersection of the two segments represented the early component onset time.

The peak amplitude was measured as the maximum amplitude of the potential occurring around the time of the cue. In movement-related conditions, this measure reflected the contribution of the late component of the MRP. Measures of the early slope and peak amplitude across all four conditions were analysed by one-way repeated-measures ANOVA, with pairwise comparisons between conditions analysed by Tukey tests. The onset time could only be measured for conditions in which an early pre-movement rise in activity was observed and was subsequently analysed by a dependent *t*-test comparing performing and imagining movement conditions.

MRPs for imagined movements were further analysed by separating the two counter-balancing groups. Potentials for subjects who performed the moving condition first, and therefore had practice at performing movement before imagining (with practice), were compared with subjects who had performed the imagining condition first and therefore had no such practice (without practice). The early slopes of potentials for these two groups were analysed by an independent *t*-test.

In the localisation phase of the study, differences in the scalp distributions of early and late components of MRPs when imagining compared with actually performing movement were examined. Lateralised differences in peak amplitudes were analysed by dependent *t*-tests, and early slopes and onset times were analysed by

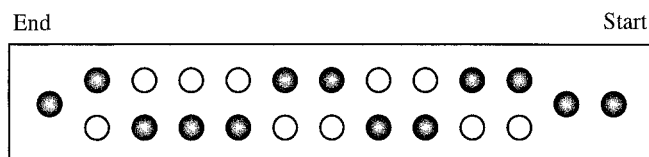
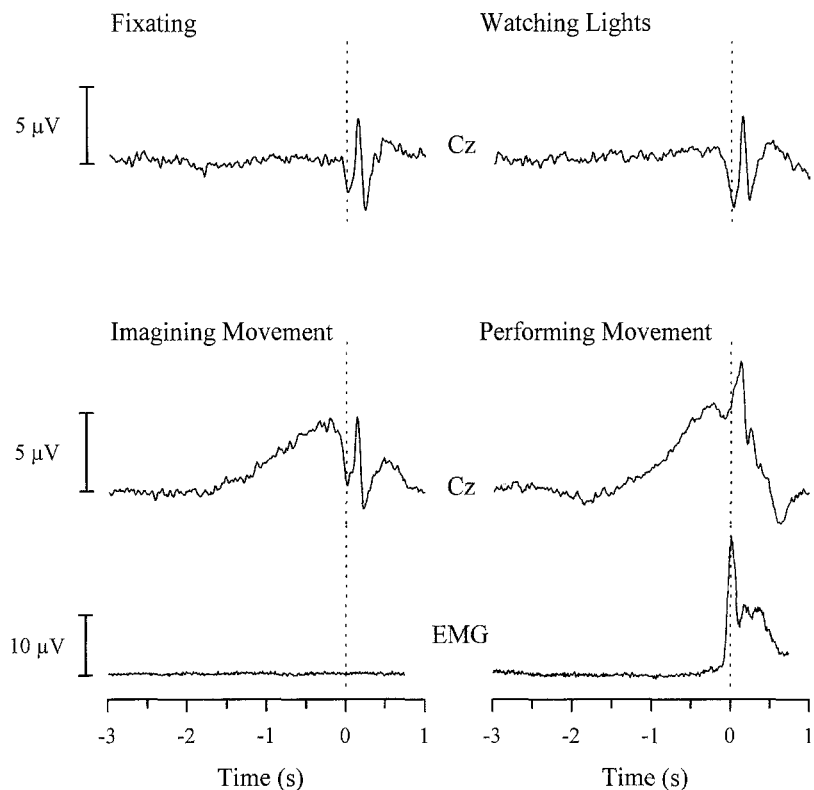


Fig. 1 The tapping board. Shaded circles represent the pathway used in the study

Fig. 2 Mean movement-related potentials from position Cz for all subjects in each experimental condition, and mean EMG responses for Imagining and Performing movement conditions. Potentials are shown from 3 s before to 1 s after the cue, with the dotted line marking the time of presentation of the cue



two-way ANOVA, with repeated-measures factors of Condition (Imagining compared with Performing movement) and scalp location (nine recording sites).

Results

Mean event-related potentials recorded from position Cz for all subjects are shown in Fig. 2. As can be seen, there appears to be little, if any, early component rise in activity prior to the cue for non-movement-related conditions (watching lights and fixating on centre); however, potentials for Performing movement and Imagining movement conditions both show an early rise in pre-movement activity of apparently similar magnitude and beginning at a similar time prior to the cue. This is most apparent in measurement of the early slope (Fig. 3a) and early component onset time (Fig. 3b).

Single-sample *t*-tests showed that for both Imagining and Performing movement conditions a significantly increased early slope was present ($t_{10}=5.02$, $P<0.001$; and $t_{10}=5.34$, $P<0.001$, respectively); however, the early slopes for both watching lights and fixating conditions were not significantly different from zero ($t_{10}=1.27$, $P>0.05$; and $t_{10}=1.52$, $P>0.05$, respectively).

ANOVA showed the early slope to vary significantly between conditions overall, $F_{3,30}=12.55$, $P<0.001$. Tukey pairwise comparisons showed the early slopes of potentials for both movement-related conditions (Performing movement and Imagining movement) to be significantly greater ($P<0.01$) than for non-movement-related conditions (watching lights and fixating on centre). Further,

there was no significant difference ($P>0.05$) in early slopes between Imagining and Performing movement conditions. Similarly, a dependent *t*-test showed no significant difference in early component onset times between Imagining and Performing movement conditions, $t(10)=0.45$, $P>0.05$. Therefore, the early component of the potential does not significantly differ in amplitude or onset time when imagining compared with actually performing movement.

In contrast, averaged potentials for Imagining and Performing movement conditions appear to differ around the time of the cue, during the late component of the MRP (Fig. 2). The amplitude of the potential appears greatest in the Performing movement condition, as indicated in measurement of the peak amplitude (Fig. 3c). ANOVA showed the peak amplitude to vary significantly between conditions overall, $F_{3,30}=7.71$, $P<0.001$. Tukey comparisons showed the peak amplitude to be significantly greater in the Performing movement condition than all other conditions ($P<0.05$), but found no significant difference in peak amplitudes between fixating, watching and imagining movement conditions ($P>0.05$). The late component of the potential is therefore significantly greater when actually performing compared with imagining movement.

Mean potentials reflecting EMG activation of the right arm biceps muscle for Performing movement and Imagining movement conditions are also shown in Fig. 2. As would be expected, a large EMG response was observed during the Performing movement condition at the time of the cue, indicating movement of the right arm; however, no such EMG response was observed dur-

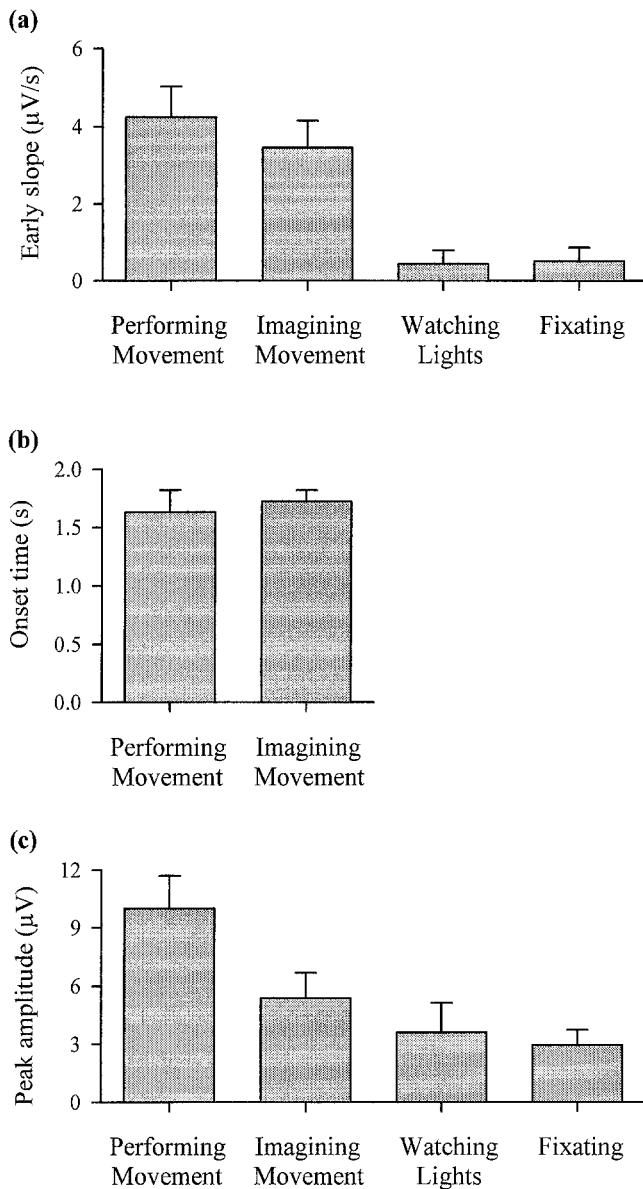


Fig. 3 Means and standard errors of the early slope (a), early component onset time (b) and peak amplitude (c) of movement-related potentials from position Cz for all subjects in each experimental condition

Fig. 4 Mean movement-related potentials from position Cz associated with imagined movements for subjects with and without practice at performing movement before imagining. Potentials are shown from 3 s before to 1 s after the cue, with the dotted line marking the time of presentation of the cue

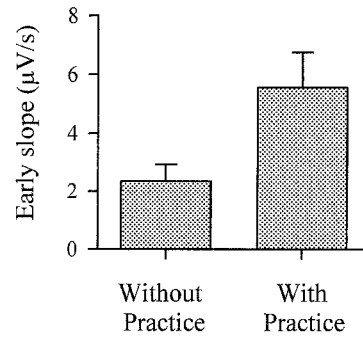
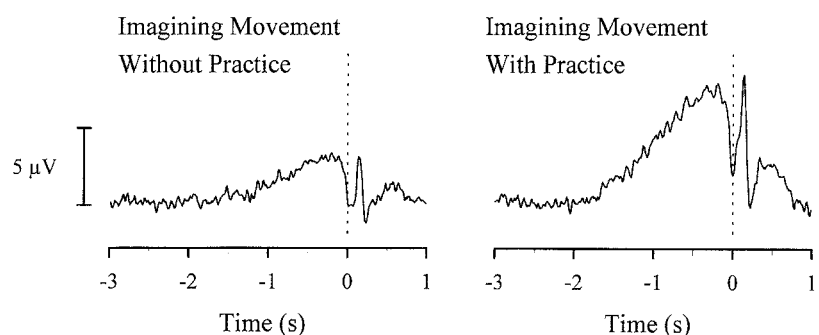


Fig. 5 Means and standard errors of the early slope of movement-related potentials from position Cz associated with imagined movements for subjects with and without practice at performing movement before imagining

ing the Imagining movement condition, indicating no muscle activation or actual movement during the task. Therefore, conditions of Performing and Imagining movement did indeed differ in that the motor imagery task involved no movement execution, nor covert EMG responses.

Mean MRPs associated with imagined movements for subjects with practice at performing movement before imagining, compared with those without practice, are shown in Fig. 4. As can be seen, the early component of the MRP appears to be greatly reduced in subjects without practice at performing movement before imagining movement. This difference is reflected in measurement of the early slope (Fig. 5). An independent *t*-test showed the early slope to be significantly greater when subjects had prior practice at performing movement before imagining, than with no such practice, $t_{10}=2.40$, $P<0.05$; however, the early slope of potentials for subjects without practice was still significantly greater than zero, $t_5=3.95$, $P<0.05$.

Mean event-related potentials across scalp locations for conditions of Performing movement, Imagining movement and watching lights are shown in Fig. 6. For both movement-related conditions (Performing and Imagining movement), an early pre-movement rise in activity was clearly present and was examined in more detail. For both Performing and Imagining movement conditions, the early slope (Table 1) was greatest at position Cz (near the supplementary motor area), followed by position C3 (near the left-side, right-hand primary motor

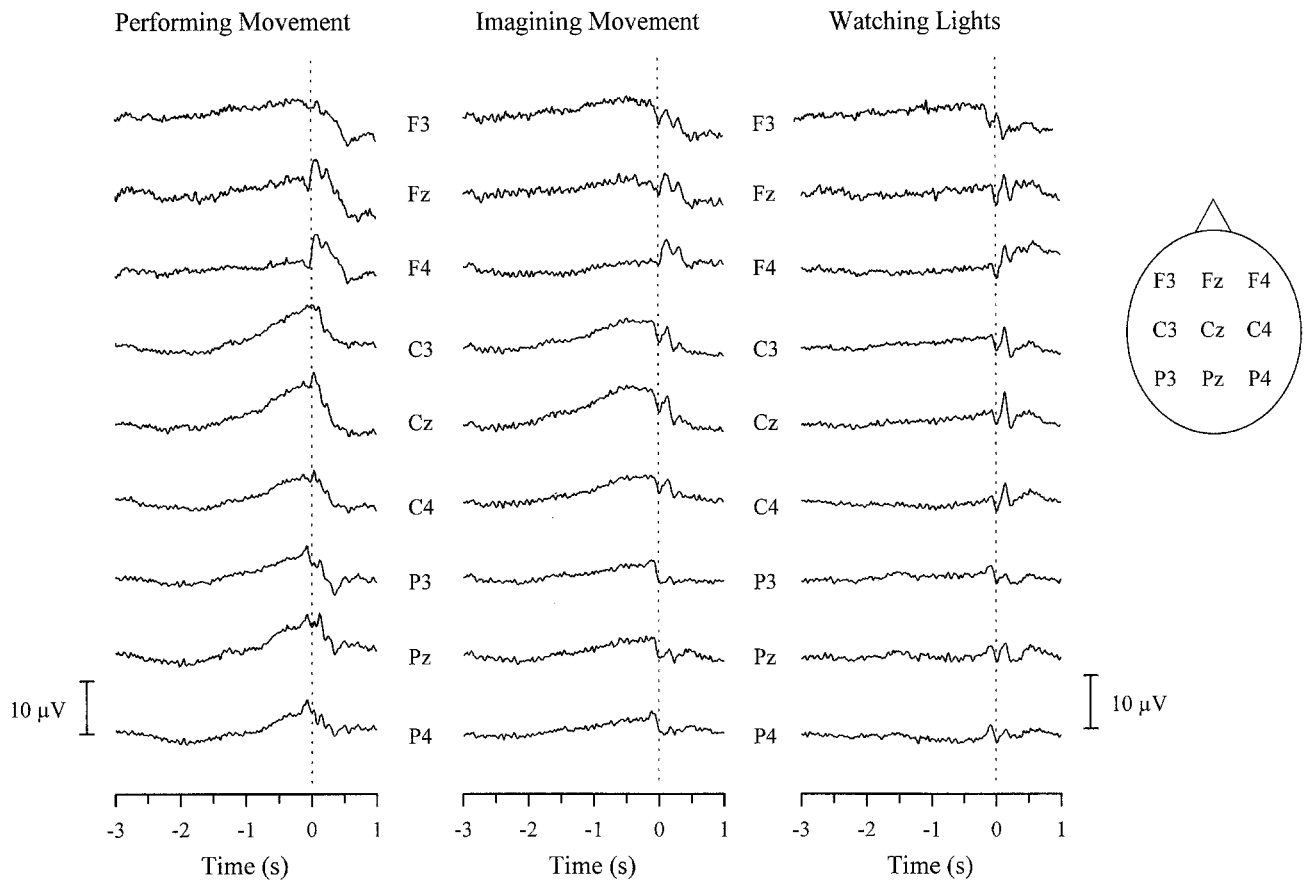


Fig. 6 Mean movement-related potentials across scalp locations for each experimental condition. Potentials are shown from 3 s before to 1 s after the cue, with the *dotted line* marking the time of presentation of the cue

Table 1 Means of the early slope, early component onset time and peak amplitude of movement-related potentials across scalp locations for conditions of Imagining and Performing movement

Condition	F3	Fz	F4	C3	Cz	C4	P3	Pz	P4
Early slope ($\mu\text{V/s}$)									
Imagining movement	2.64	2.54	2.41	4.45	5.55 ^a	3.98	1.97	2.76	1.99
Performing movement	1.62	2.44	0.49	4.78	4.84 ^a	3.59	3.01	3.75	2.90
Onset time (ms prior to cue)									
Imagining movement	1.46	1.33	1.38	1.42	2.00 ^a	1.36	1.72	1.60	1.58
Performing movement	1.85	1.85	1.30	1.89	1.90 ^a	1.75	1.50	1.36	1.08
Peak amplitude (μV)									
Imagining movement	5.04	6.79	7.46	7.20	8.22 ^a	7.40	5.16	6.57	6.22
Performing movement	6.69	9.59	9.25	12.26 ^a	10.77	8.74	7.71	11.49	8.56

^a The greatest value in each condition

area). Similarly, the onset time (Table 1), indicating the earliest onset of the MRP, was greatest in potentials recorded at position Cz for both Performing and Imagining movement conditions.

ANOVA showed that there was a significant difference in early slopes of potentials distributed over different scalp locations, $F_{8,40}=4.52$, $P<0.001$; however, there was no significant difference between Performing and Imagining movement conditions, $F_{1,5}=0.73$, $P>0.05$, nor was there a significant interaction, $F_{8,40}=1.43$, $P>0.05$. Similarly, the early component onset time was found to differ significantly across scalp locations, $F_{8,40}=2.27$,

$P<0.05$; however, there was no significant difference between conditions, $F_{1,5}=0.26$, $P>0.05$, nor significant interaction, $F_{8,40}=1.87$, $P>0.05$. This indicates that topographic characteristics of the early component of MRPs do not significantly differ when imagining compared with actually performing movement.

Topographic differences in the late component of MRPs were also examined by comparing peak amplitudes across scalp locations (Table 1). When performing movement, the peak amplitude was greatest at position C3 (closest to the right hand area of primary motor cortex), and was significantly greater than that at position

C4 (the equivalent site on the opposite hemisphere), $t_5=2.61$, $P<0.05$. However, when imagining movement, peak amplitudes at positions C3 and C4 were not significantly different ($t_5=0.30$, $P>0.05$), indicating a bilaterally symmetrical distribution of the late component.

Discussion

When imagining movement, averaged potentials recorded at the vertex revealed an early pre-movement rise in activity similar to that observed prior to actual movement. The early slope and onset time of the early potential preceding imagined movement was not significantly different to that when actually performing movement. This is consistent with the suggestion that the early component of the MRP reflects motor preparatory processes which may operate both when imagining and actually performing movement.

In contrast, the late component of the MRP, reflected in measurement of the peak amplitude, was significantly greater when performing compared with imagining movement. Further, the late component was lateralised, with the greatest amplitude over the right hand area of primary motor cortex when performing movement, but not when imagining movement. This suggests that the late component of the MRP reflects mainly activity of the primary motor cortex associated with movement execution which may only operate during actual movement performance.

The localisation phase of the study showed a significant difference between early slopes and onset times of potentials at different recording sites. This indicates that the early component of the MRP showed a specific topographic distribution across scalp locations, with the greatest amplitude and earliest onset at position Cz, near the supplementary motor area. The lack of significant difference between Performing and Imagining movement conditions, and the lack of significant interaction, indicates that the topographic characteristics of the early component of the potential do not significantly differ between these conditions. Therefore, the early component of the potential most probably reflects similar pre-movement processes generated at the same cortical site when imagining compared with actually performing movement.

For non-movement-related conditions (watching lights and fixating), no significant early rise in activity was observed. The fixating condition reveals components owing to visual evoked potentials relating to the extinction of light cues, while the condition of watching lights also reveals components of the potential owing to horizontal eye movements during the task. The absence of early pre-cue potentials during these conditions therefore indicates that the regular extinction of light cues and eye movements during performance of the task do not significantly influence the early component of movement-related potentials during movement performance.

Similarly, the absence of increasing pre-cue activity for non-movement-related conditions indicates the ab-

sence of cognitive potentials, such as the CNV, related to expectancy or anticipation of the regular cue. Conditions of imagining movement and watching lights both involved identical regular cues requiring similar attentional and anticipatory mechanisms. If increased pre-cue activity was related to the CNV, the early component of potentials would be similar for conditions of imagining movement and watching lights; however, this was not the case. Increased pre-cue activity was only observed for conditions in which a motor response, whether real or imagined, was required; therefore, such activity most probably reflects motor preparatory processes, rather than expectancy or anticipation of the regular cue.

Previous studies of regional cerebral blood flow clearly indicate that the supplementary motor area, and not primary motor area, is significantly activated during the mental rehearsal and imagination of sequential movement (Roland et al. 1980; Decety et al. 1988; Rao et al. 1993). Therefore, in this study, cortical activity reflected in potentials associated with imagined movement most likely reflects increasing activity of the supplementary motor area, with little contribution from primary motor areas. Further, no significant differences exist between spatial and temporal characteristics of the early component of potentials when imagining compared with actually performing movement. Therefore, the early component of potentials associated with movement *performance* also most probably reflects increasing activity arising predominantly from the supplementary motor area.

Results also showed an effect of learning on the early component of potentials associated with imagined movements. The early slope of the potential was greater for subjects with practice at performing movement before imagining. Previous studies have shown that the supplementary motor area is most involved in the performance of well-learned movements (see Cunnington et al., in press, for review), with activity of the supplementary motor area increasing as novel tasks become well-learned with practice (Grafton et al. 1992). Results of this study may therefore reflect properties of supplementary motor area function, since cortical activity was not as great without practice when the imagery task was novel, but increased following practice in performing movement when subjects may have learnt strategies required for later imagery of the same movement. This finding is consistent with and further supports the conclusion that the early component of MRPs reflects mainly increasing activity of the supplementary motor area prior to movement.

In conclusion, therefore, the early component of the MRP most probably reflects preparatory processes directly related to movement, rather than simply attention or anticipation of a forthcoming response. Further, the supplementary motor area is most likely the generator of the early component of the MRP, since pre-movement activity prior to imagined and executed movements did not significantly differ, and previous studies have shown only the supplementary motor area to be active during

motor imagery. Therefore, the early component of the MRP most probably reflects pre-movement preparatory activity arising mainly from the supplementary motor area, which is involved prior to both movement execution and motor imagery.

References

- Beisteiner R, Höllinger P, Lindinger G, Lang W, Berthoz A (1995) Mental representations of movements. Brain potentials associated with imagination of hand movements. *Electroencephalogr Clin Neurophysiol* 96:183–193
- Benecke R, Dick JPR, Rothwell JC, Day BL, Marsden CD (1985) Increase of the Bereitschaftspotential in simultaneous and sequential movements. *Neurosci Lett* 62:347–352
- Böcker KBE, Brunia CHM, Cluitmans PJM (1994) A spatio-temporal dipole model of the readiness potential in humans. I. Finger movement. *Electroencephalogr Clin Neurophysiol* 91:275–285
- Bötzel K, Plendl H, Paulus W, Scherg M (1993) Bereitschaftspotential: is there a contribution of the supplementary motor area? *Electroencephalogr Clin Neurophysiol* 89:187–196
- Ceballos-Baumann AO, Marsden CD, Passingham RE, Stephan KM, Frackowiak RSJ, Brooks DJ (1994) Cerebral activation with performing and imagining movement in idiopathic torsion dystonia (ITD): a PET study (abstract). *Neurology [Suppl 2]* 44:A338
- Cunnington R, Bradshaw JL, Ianssek R (in press) The role of the supplementary motor area in the control of voluntary movement. *Hum Mov Sci*
- Cunnington R, Ianssek R, Bradshaw JL, Phillips JG (1995) Movement-related potentials in Parkinson's disease: presence and predictability of temporal and spatial cues. *Brain* 118:935–950
- Decety J, Michel F (1989) Comparative analysis of actual and mental movement times in two graphic tasks. *Brain Cogn* 11:87–97
- Decety J, Philippon B, Ingvar DH (1988) rCBF landscapes during motor performance and motor ideation of a graphic gesture. *Eur Arch Psychiatry Neurol Sci* 238:33–38
- Decety J, Jeannerod M, Germain M, Pastene J (1991) Vegetative response during imagined movement is proportional to mental effort. *Behav Brain Res* 42:1–5
- Decety J, Jeannerod M, Durozard D, Baverel G (1993) Central activation of autonomic effectors during mental simulation of motor actions in man. *J Physiol* 461:549–563
- Deecke L (1987) Bereitschaftspotential as an indicator of movement preparation in supplementary motor area and motor cortex. In: Porter R (ed) *Motor areas of the cerebral cortex*. (Ciba Foundation Symposium 132) Wiley, Chichester, pp 231–250
- Deecke L, Scheid P, Kornhuber HH (1969) Distribution of readiness potential, pre-motion positivity, and motor potential of the human cerebral cortex preceding voluntary finger movements. *Exp Brain Res* 7:158–168
- Deiber M-P, Passingham RE, Colebatch JG, Friston KJ, Nixon PD, Frackowiak RSJ (1991) Cortical areas and the selection of movement: a study with positron emission tomography. *Exp Brain Res* 84:393–402
- Grafton ST, Mazziotta JC, Presty S, Friston KJ, Frackowiak RS, Phelps ME (1992) Functional anatomy of human procedural learning determined with regional cerebral blood flow and PET. *J Neurosci* 12:2542–2548
- Ikeda A, Luders HO, Burgess C, Shibasaki H (1992) Movement-related potentials recorded from supplementary motor area and primary motor area: role of supplementary motor area in voluntary movements. *Brain* 115:1017–1043
- Lang W, Lang M, Uhl F, Koska C, Kornhuber A, Deecke L (1988) Negative cortical DC shifts preceding and accompanying simultaneous and sequential finger movements. *Exp Brain Res* 71:579–587
- Lassen NA, Ingvar DH, Skinhøj E (1978) Brain function and blood flow. *Sci Am* 239(4):50–59
- Mattingley JB, Bradshaw JL, Phillips JG (1992) Impairments of movement initiation and execution in unilateral neglect: directional hypokinesia and bradykinesia. *Brain* 115:1849–1874
- Naito E, Matsumura M (1994) Movement-related slow potentials during motor imagery and motor suppression in humans. *Brain Res Cogn Brain Res* 2:131–137
- Neshige R, Luders H, Shibasaki H (1988) Recording of movement-related potentials from scalp and cortex in man. *Brain* 111:719–736
- Rao SM, Binder JR, Bandettini PA, Hammeke TA, Yetkin FZ, Jesmanowicz A, Lisk LM, Morris GL, Mueller WM, Estkowski LD, Wong EC, Haughton VM, Hyde JS (1993) Functional magnetic resonance imaging of complex human movements. *Neurology* 43:2311–2318
- Roland PE, Larsen B, Lassen NA, Skinhøj E (1980) Supplementary motor area and other cortical areas in organization of voluntary movements in man. *J Neurophysiol* 43:118–136
- Romo R, Schultz W (1992) Role of primate basal ganglia and frontal cortex in the internal generation of movements. III. Neuronal activity in the supplementary motor area. *Exp Brain Res* 91:396–407
- Shibasaki H, Barrett G, Halliday E, Halliday AM (1980) Components of the movement-related cortical potential and their scalp topography. *Electroencephalogr Clin Neurophysiol* 49:213–226
- Shibasaki H, Sadato N, Lyshkow H, Yonekura Y, Honda M, Nagamine T, Suwazono S, Magata Y, Ikeda A, Miyazaki M, Fukuyama H, Asato R, Konishi J (1993) Both primary motor cortex and supplementary motor area play an important role in complex finger movement. *Brain* 116:1387–1398
- Stephan KM, Fink GR, Passingham RE, Silbersweig D, Ceballos-Baumann AO, Frith CD, Frackowiak RSJ (1995) Functional anatomy of the mental representation of upper extremity movements in healthy subjects. *J Neurophysiol* 73:373–386
- Tamas LB, Shibasaki H (1985) Cortical potentials associated with movement: a review. *J Clin Neurophysiol* 2:157–171
- Tanji J, Shima K (1994) Role for supplementary motor area cells in planning several movements ahead. *Nature* 371:413–416
- Tarkka IM (1994) Electrical source localisation of human movement-related cortical potentials. *Int J Psychophysiol* 16:81–88
- Toro C, Matsumoto J, Deuschl G, Roth BJ, Hallett M (1993) Source analysis of scalp-recorded movement-related electrical potentials. *Electroencephalogr Clin Neurophysiol* 86:167–175
- Walter WG, Cooper R, Aldridge VJ, McCallum WC, Winter AL (1964) Contingent negative variation: an electrical sign of sensorimotor association and expectancy in the human brain. *Nature* 203:380–384